

Marvin Joseph B
Senior Production SQL Server DBA | AI & Generative AI Engineering
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PROFESSIONAL SUMMARY

Senior Production SQL Server DBA with 6+ years of experience supporting mission-critical production databases, incident response, automation, high availability, cloud migrations, and deployment workflows, now transitioning into **AI and Generative AI engineering**.

Builds agentic AI applications with **Python, FastAPI, Pydantic, LLM APIs, structured outputs, tool calling, RAG, evaluation, Docker, and CI/CD**, applying hands-on production troubleshooting experience to AI systems designed for real operational workflows.

TECHNICAL SKILLS

Generative AI & Agentic Systems: LLM APIs, Tool / Function Calling, Agentic Workflows, Agent Orchestration, Structured Outputs, RAG, Embeddings, Vector Search, Prompt Engineering, Context Management, Human-in-the-Loop, AI Evaluation

Backend Engineering: Python, FastAPI, Pydantic, REST APIs, JSON, API Integration, Async Patterns, Error Handling, Retries, Timeouts

AI Reliability & Safety: Output Validation, Tool Permission Boundaries, Human Approval Gates, Least-Privilege Access, Audit Logging, Secrets Management, Observability, Failure Handling, Prompt Injection Awareness

DevOps & Infrastructure: Docker, GitHub Actions, Git, GitHub, CI/CD, Linux, Nginx, Azure DevOps, Octopus Deploy, VPS Deployment

Database & Data: SQL Server 2012–2022, Azure SQL Managed Instance, Azure SQL Database, Always On Availability Groups, Query Store, Execution Plans, DMVs, Wait Statistics, Blocking, Deadlocks, TempDB, HA/DR, Backup & Recovery, T-SQL, PowerShell, Pandas, SSIS, CDC, ETL

SELECTED AI ENGINEERING PROJECT

Incident Investigation Agent

Python | FastAPI | Pydantic | LLM Tool Calling | Agentic Workflows | Docker | GitHub Actions

Built an agentic AI system inspired by hands-on experience investigating production database and application incidents.

Architecture:

Incident → Orchestrator → Diagnostic Tools → Evidence → Analysis → Root-Cause Hypotheses → Recommended Actions → Human Approval

- Built an **agentic investigation workflow** in which the LLM determines required evidence and invokes narrowly scoped diagnostic tools instead of receiving unrestricted production access.
 - Implemented diagnostic tool interfaces for retrieving **application errors, database wait statistics, blocked sessions, database health, connection metrics, deployment history, operational runbooks, and previous incident evidence**.
 - Built the backend with **Python, FastAPI, and Pydantic**, using structured schemas to control communication between orchestration logic, diagnostic tools, and LLM outputs.
 - Implemented structured outputs for **incident timelines, observed evidence, root-cause hypotheses, supporting and conflicting evidence, uncertainty, and recommended actions**, while separating observed facts from model-generated conclusions.
 - Incorporated **least-privilege tool access and human approval gates** so the agent can investigate autonomously without unrestricted authority to execute destructive production actions.
 - Built evaluation scenarios covering **connection exhaustion, blocking and lock contention, query-plan regression, database performance degradation, and post-deployment application failures**, with retries, timeouts, schema validation, structured logging, automated testing, Docker, and CI/CD.
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PROFESSIONAL EXPERIENCE

Diplomatic Solutions Corporation — Frederick, MD

Senior SQL Server DBA | September 2021 – Present

Support 24x7 production SQL Server environments spanning **20+ SQL Server instances, 100+ databases, multi-terabyte transactional workloads, and databases exceeding 2 TB**, while maintaining **99.95%+ database availability year over year**.

- Serve as an escalation point for **P1/P2 production incidents**, restoring services within SLA in **95%+ of cases** while investigating database failures, application/database connectivity problems, performance degradation, failed deployments, and high-availability events.
- Diagnose complex issues involving **blocking, deadlocks, connection exhaustion, query-plan regressions, CPU and I/O pressure, TempDB contention, and application connectivity**, contributing to a **40–60% reduction in recurring incidents** through root-cause remediation.

- Correlate evidence from **DMVs, Query Store, execution plans, wait statistics, Extended Events, monitoring platforms, application behavior, deployment history, and runbooks** to isolate root causes and prevent repeat incidents.
 - Perform structured incident investigations involving **evidence collection, timeline reconstruction, hypothesis development, validation, remediation, and root-cause analysis** across business-critical production systems.
 - Build **Python, PowerShell, and T-SQL automation** for health checks, diagnostics, monitoring, maintenance, and repetitive production operations, reducing manual DBA workload by approximately **50%**.
 - Develop and maintain operational **runbooks** covering troubleshooting procedures, recovery actions, known failure patterns, escalation paths, and production safeguards.
 - Support application and database deployment workflows using **GitHub, Azure DevOps, CI/CD pipelines, and Octopus Deploy**, including investigation and remediation of production regressions following releases.
 - Establish performance baselines and troubleshoot abnormal workloads using **Query Store, execution plans, indexing, statistics, wait analysis, and SQL Server diagnostic tooling**.
 - Administer **Always On Availability Groups and Windows Failover Clustering**, executing planned and unplanned failovers with **RTO under 5 minutes for critical systems**.
 - Maintain backup and transaction-log recovery strategies that achieved **100% successful restore verification during quarterly recovery testing**, including recovery support during production incidents.
 - Support migrations to **Azure SQL Managed Instance**, including compatibility validation, migration execution, data verification, application connectivity, and near-zero-downtime cutover support.
 - Standardize and optimize **100+ SQL Server Agent jobs** supporting maintenance, monitoring, scheduling, and operational automation.
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EARLIER EXPERIENCE

Emitek — Lanham, MD

ETL Engineer / SQL Server DBA | July 2020 – August 2021

- Developed data-processing and automation workflows using **Python, Pandas, SQL Server, APIs, and SSIS**.
- Built ETL pipelines integrating structured and semi-structured data from **SQL Server, flat files, XML, and external APIs**.
- Implemented **incremental loading and Change Data Capture (CDC)** to improve processing efficiency and data freshness.
- Developed Python-based transformations and automation for requirements beyond traditional SSIS workflows.

- Investigated production **ETL failures, data inconsistencies, query-performance regressions, and database bottlenecks.**
 - Optimized SQL queries, indexes, and execution plans and automated scheduling, monitoring, and failure handling using **SQL Server Agent.**
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EDUCATION

Bachelor of Science, Information Technology
Western Governors University
2026

Associate Degree, Computer Science / Information Systems
Montgomery College — 2021

CERTIFICATIONS

Claude Certified Associate — Foundations — Anthropic

CompTIA Security+

Additional Training: SQL Server Database Administration | Windows Server System Administration